

1. General Description

The EMS3218 is a high performance, Single-Pole, Single-Throw (SPST) multiplexer that combines a low-distortion audio switch path. The architecture is designed to allow audio signals to swing below ground. The EMS3218 includes an over voltage protection of Ro and Lo pins to protect internal port. Typical applications involve switching in portables and consumer applications such as mobile phone, tablet, laptop, portable media players. The EMS3218 is available in QFN 1.4x1.8-10L package. Standard products are Pb-Free and halogen-Free.

2. Features and Benefits

- Single supply range: 2.5 V to 4.5 V
- MUTE pin voltage: 17 V
- Ro and Lo pins OVP threshold: 4.7 V
- Audio path On-Resistance: 0.78 Ω
- THD+N @ 32 Ω load & 0.7 Vrms: -105 dB
- Signal-to-Noise (SNR) Ratio: -115 dB
- Audio Swing Range ($R_L = 1K\Omega$): -2.5 V to 3 V

3. Applications

- HiFi Audio Equipment

EMS3218

HiFi Audio Switches with Negative Swing Capability and Over-Voltage-Protection

4. Ordering Information

Table 1. Ordering information

Type number	Topside marking	Package		
		Name	Description	Quantity
EMS3218DRB	218	QFN 1.4 x 1.8 - 10L	QFN package, 10 pins 1.4 mm x 1.8 mm; 0.55 mm (Max) height	3000

5. Function Diagram

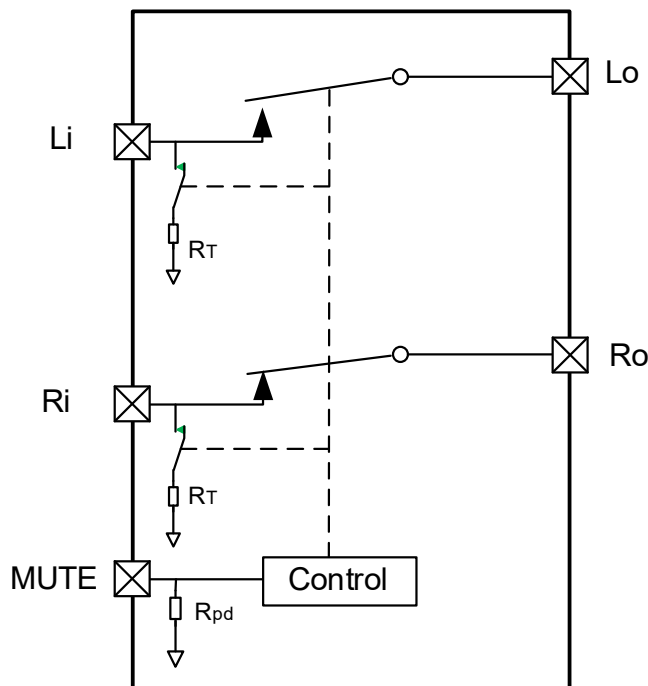


Fig 1. Function Diagram

[1] 70Ω R_T resistors are enabled when MUTE=H.

6. Pinning Information

6.1. Pin map

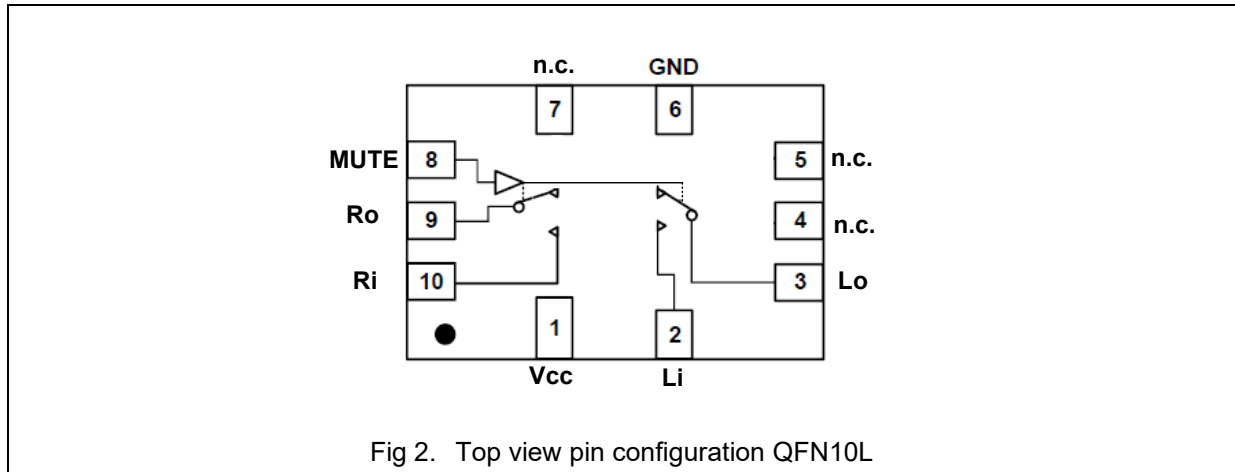


Fig 2. Top view pin configuration QFN10L

6.2. Pin description

Table 2. Pin description

Symbol	Pin	Description
V _{CC}	1	Power Supply
Li	2	Audio Left Input Source
Lo	3	Audio Left Output
n.c.	4,5,7	Not Connection
GND	6	Ground
MUTE	8	When MUTE = 1, Audio Paths are OFF; When MUTE = 0, Audio Paths are ON
Ro	9	Audio Right Output
Ri	10	Audio Right Input Source

7. Functional Description

Table 3. Function table

H = HIGH voltage level; L = LOW voltage level.

MUTE	Status of Lo and Ro Status
H	Lo disconnected to Li, Ro disconnected to Ri
L	Lo connected to Li, Ro connected to Ri

8. Absolute Maximum Ratings

Stresses exceeding the absolute maximum ratings may damage the device. The device may not function or be operable above the recommended operating conditions and stressing the parts to these levels is not recommended. In addition, extended exposure to stresses above the recommended operating conditions may affect device reliability. The absolute maximum ratings are stress ratings only.

Table 4. Absolute Maximum Ratings

In accordance with the Absolute Maximum Rating System (IEC 60134). Voltages are referenced to GND.

Symbol	Parameter	Conditions	Min	Max	Unit
V _{CC}	Supply voltage range		- 0.3	6.5	V
V _{ADUDIO}	Switch signal voltage range		- 3	V _{CC}	V
V _{VBUS}	MUTE pin voltage range		- 0.3	18	V
I _{OUT}	Continues output current		-200	200	mA
T _J	Junction temperature range			150	°C
T _L	Lead temperature range			260	°C
T _{stg}	Storage temperature range		- 65	150	°C
R _{θJA}	Thermal resistance			250	°C/W
ESD	ESD Capability	MUTE to GND	± 4		kV
		V _{CC} to GND	± 5		kV
		Lo and Ro to GND	± 4		kV
		Li and Ri to GND	± 4		kV

9. Recommended Operating Conditions

The Recommended Operating Conditions table defines the conditions for actual device operation. Recommended operating conditions are specified to ensure optimal performance to the datasheet specifications. EnergyMath does not recommend exceeding them or designing to Absolute Maximum Ratings.

Table 5. Recommended Operating Conditions

Symbol	Parameter	Conditions	Min	Max	Unit
V _{CC}	Supply voltage range		2.5	4.5	V
V _{audio}	Audio signal voltage range		-2.5	3.0	V
V _{VBUS}	MUTE pin voltage range		0	16	V
T _{OPR}	Operating temperature range		- 40	85	°C

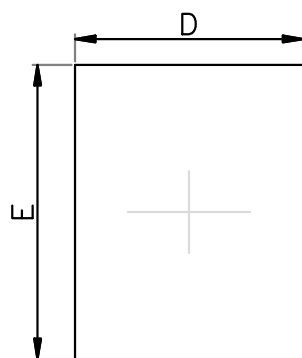
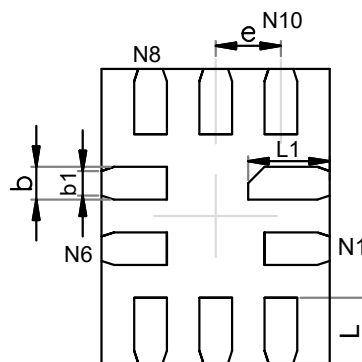
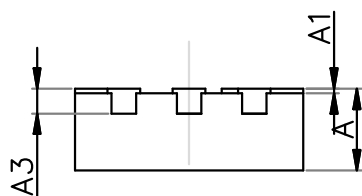
10. Electrical Characteristics

Table 6. Static characteristics

Unless otherwise specified, typical values are at $V_{CC} = 3.3\text{ V}$, $T_A = +25\text{ }^{\circ}\text{C}$.

Symbol	Parameter	Conditions	Min	Typ	Max	Unit
DC Electrical Characteristics						
I_{CC}	Quiescent supply current			54		μA
I_{IN}	MUTE input leakage current	$0 < V_{IS} < 17\text{ V}$		3.5		μA
I_{OFF}	Switch off leakage current			34		μA
R_{ON}	Switch on resistance	$I_{SW} = 60\text{ mA}$, $V_{SW} = -2\text{ V to } 2\text{ V}$		0.78		Ω
ΔR_{ON}	Switch delta R_{ON}	$I_{SW} = 60\text{ mA}$, $V_{SW} = -2\text{ V to } 2\text{ V}$		50		$\text{m}\Omega$
$R_{FLAT(ON)}$	Switch R_{ON} flatness	$I_{SW} = 60\text{ mA}$, $V_{SW} = -2\text{ V to } 2\text{ V}$		75		$\text{m}\Omega$
C_{ON}	On capacitance	$V_{Bias} = 0.2\text{ V}$, $f = 1\text{ MHz}$		5		pF
C_{OFF}	Off capacitance	$V_{Bias} = 0.2\text{ V}$, $f = 1\text{ MHz}$		4		pF
AC Electronics Characteristics						
X_{talk}	Non-Adjacent channel crosstalk	$f=1\text{ KHz}$, $R_T = 32\text{ }\Omega$		- 90		dB
THD	Total harmonic distortion	$f = 20\text{ Hz to } 20\text{ KHz}$, $R_L = 32\text{ }\Omega$, $V_{IN} = 0.7\text{ V}_{rms}$		- 105		dB
		$f = 20\text{ Hz to } 20\text{ KHz}$, $R_L = 1\text{ k}\Omega$, $V_{IN} = 1.5\text{ V}_{rms}$		-110		dB
O_{IRR}	Off isolation rejection ratio	$f=1\text{ KHz}$, $R_T = 32\text{ }\Omega$		- 110		dB
SNR	Signal-to-Noise ratio	$f = 20\text{ Hz to } 20\text{ KHz}$, $R_L = 32\text{ }\Omega$, $V_{IN} = 2\text{ V}_{rms}$		-115		MHz
V_{OVP}	OVP Lockout Threshold	R_o and L_o pins OVP threshold		4.7		V
Control logic						
R_{PD}	MUTE internal Pull-Down resistors			4		$\text{M}\Omega$
R_T	Audio path termination resistors			70		Ω
V_{IH}	Control input voltage HIGH	$2.5\text{ V} < V_{CC} < 3.0\text{ V}$	1.3			V
		$3.0\text{ V} < V_{CC} < 3.5\text{ V}$	1.5			V
V_{IL}	Control input voltage LOW	$2.5\text{ V} < V_{CC} < 3.0\text{ V}$			0.3	V
		$3.0\text{ V} < V_{CC} < 3.5\text{ V}$			0.5	V

11. Package Outline

DFN 1.4x1.8 -10L

TOP VIEW

BOTTOM VIEW

SIDE VIEW

Symbol	Dimensions In Millimeters		Dimensions In Inches	
	Min.	Max.	Min.	Max.
A	0.450	0.550	0.018	0.022
A1	0.000	0.050	0.000	0.002
A3	0.152REF.		0.006REF.	
D	1.350	1.450	0.053	0.057
E	1.750	1.850	0.069	0.073
b	0.150	0.250	0.006	0.010
b1	0.100	0.200	0.004	0.008
e	0.400TYP.		0.016TYP.	
L	0.350	0.450	0.014	0.018
L1	0.450	0.550	0.018	0.022

12. Tape and Reel Information

12.1. Carrier tape dimensions

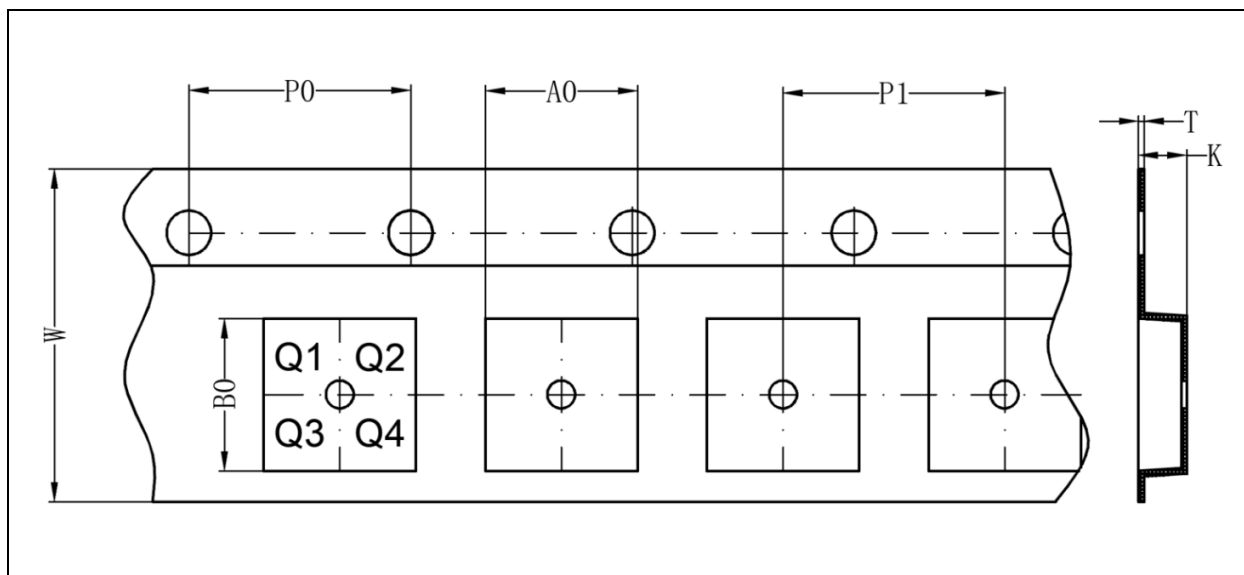


Table 7. Carrier tape dimensions

Package version	A0(mm)	B0(mm)	K0(mm)	T(mm)	P1(mm)	W(mm)	P0(mm)	PIN 1
QFN1.4×1.8-10L	1.6	2.0	0.85	0.2	4	8	4	Q1

12.2. Reel and box dimensions

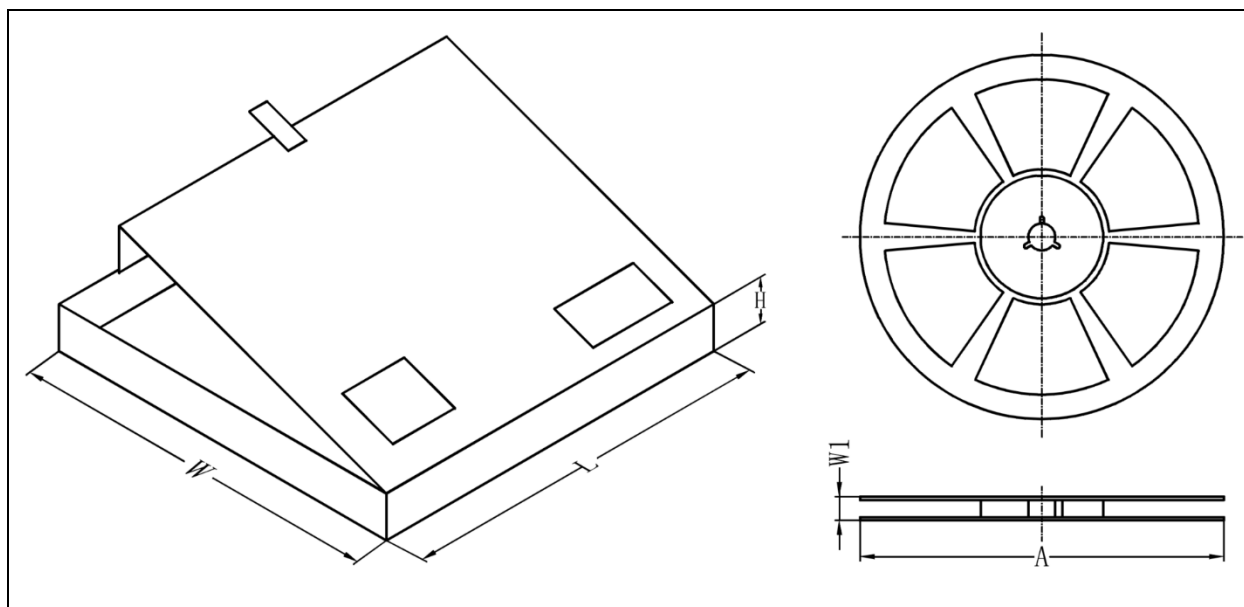


Table 8. Dimensions and quantities

Package version	Type NO. ending	Reel Dimension A (mm)	Max Reel Width W1 (mm)	SPQ (pcs)[1]	Reels per box	Outer box dimensions L×W×H(mm)[2]
QFN1.4×1.8-10L	DRB	180	13.2	3000	1	210x200x40

[1] Packing quantity dependent on specific product type. Please contact your local Energymath representative for ordering.

[2] Dimensions for reference only.

13. Revision History

Table 9. Revision history

Document ID	Release Date	Data sheet status	Change notice	Supersedes
EMS3218 Rev. 0.1	Jan 21, 2026	Draft datasheet		